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Title

What a Slice-Level Benchmark Certifies without the Pixels: An Audit of Six Public Imaging Datasets

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The journal asks for academic degrees after each name. All four authors have confirmed that none holds a completed academic degree; all four are secondary-school students. The field is therefore left empty rather than filled with a credential nobody holds.

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Use of large language models. Large language model assistance was used in two ways: in drafting and editing this manuscript, and in writing the analysis code that produces every number in the Results. The tool was **Claude (Opus 5), Anthropic**, accessed 2026-07-27 to 2026-08-14.

No value reported in the paper was generated by a language model. Every number is a deterministic output of the released code, run on a published label file, and each was re-read from the code's own output before submission. The flagship estimate was reproduced by a second implementation sharing no code with the primary one: run over its own independent holdouts under the same pre-specified protocol, the two families of estimates agree to 0.0003 AUC per slice and 0.005 per patient, and the constant predictor is exactly 0.500 in every draw of both. That reproduction, rather than any assurance about provenance, is what should carry the point, and the code is released so the check can be repeated without the authors.

Other acknowledgments. The authors thank the NYU fastMRI Initiative for releasing the datasets underlying the two audited arms that derive from them, the fastMRI Prostate T2-weighted and diffusion-weighted arms. Two further arms drawn from a fastMRI+ knee release were scored in an earlier round and are withdrawn from this submission. No individual is named, so no written permission to be acknowledged was required.

Data sharing statement

Data generated by the authors or analyzed during the study are available at: `https://github.com/sathvikloke/PhaseDx`, revision `51d3d4bbf38e879af5f13124381f63ba34181e82`, released under a Creative Commons Attribution 4.0 International license.

The first sentence above is one of the five statements the journal requires verbatim, with the citation to the data supplied as the policy directs; the revision identifier is given because the Algorithm and Code Transparency policy requires one. That revision is the commit whose code produced every value reported in this manuscript; any later commit on the same branch edits manuscript prose only and changes no reported number. The authors undertake to maintain the repository and link for the five years the policy specifies, and will mint an archival persistent identifier for the revision of record on acceptance. What follows describes the restrictions on reuse, as the policy also asks.

The archive contains the audit code, the per-benchmark output payloads, the label-table preparation scripts, the frozen-holdout re-analysis, a second implementation of the intracranial hemorrhage unit-collapse computation sharing no code with the first, and the assembler that builds the cross-study comparison. It does not contain, and cannot contain, any benchmark's pixel data. It also holds material from other parts of the larger study that this submission does not report; see the cover letter.

The label files analyzed here were generated by third parties, not by the authors, and each carries its own terms. Five of the six benchmarks are public releases obtained under their own published terms. The worked-example label files are

NYU fastMRI releases, available by application to the NYU fastMRI Initiative (`https://fastmri.med.nyu.edu`) under a data sharing agreement that permits use in academic publications but forbids onward distribution of the dataset or of files derived from it; those files are therefore not redistributed in the archive, and the code that regenerates every reported value from a reader's own authorized copy is released instead. No identifiable or private participant-level information was held by the authors. The flagship label file carries no age or sex, which is why neither is reported for it; two audited label files do carry demographic or clinical columns, and Table 4 names them.

Conflicts of interest

All four authors declare no conflicts of interest. No author has a financial interest in the subject matter of this manuscript, and no author has any affiliation with, or financial involvement in, any organization with a financial interest in it. No author holds any position, fiduciary or otherwise, with any organization that produced, funds, distributes or maintains any of the benchmarks audited here, or with any organization whose published results are used as comparators. Each author will submit a completed ICMJE Disclosure of Potential Conflicts of Interest Form.

All fields on this page are complete. The language-model access dates here, in the cover letter and in the manuscript's Acknowledgments must stay identical; if the manuscript is edited again before upload, update the final date in all three and recount the body words.

What a Slice-Level Benchmark Certifies without the Pixels: An Audit of Six Public Imaging Datasets

Article Type: Original Research

Summary Statement

One score vector computed with no access to the images ranked slices far above chance and patients at or below it on the largest benchmark audited, and the difference between the two readings was carried by how many slices a patient's stack held.

Key Points

- Over 24 independently drawn patient-disjoint holdouts, one pixel-blind score vector read area under the receiver operating characteristic curve (AUC) 0.7381 (range, 0.7350 to 0.7415) per slice and, after pre-specified mean aggregation, 0.4551 (range, 0.4424 to 0.4628) per patient, while a constant predictor read exactly 0.500 at both units in every draw.
- The patient-level reading is a stack-depth confound of the mean operator: stack depth alone reads 0.394 per patient, and with depth held fixed the reading is 0.5037 (range, 0.4979 to 0.5071), covering chance across the family and sitting on its own permutation null.
- Applied unchanged to a label file not previously opened, the same protocol reproduced the divergence on a second benchmark—0.7482 per slice against 0.5712 per patient over 24 holdouts—but not the mechanism: stack depth there reads 0.5048, so the cause is benchmark-specific even where the divergence is not.

Abbreviations

AUC = area under the receiver operating characteristic curve; CI = confidence interval; DWI = diffusion-weighted imaging; ICH = intracranial hemorrhage; SD = standard deviation.

Keywords: CT; MRI; Computer Applications—Detection/Diagnosis; Technology Assessment; Brain/Brain Stem.

Abstract

Purpose: To measure what a model given no image data reaches on volumetric benchmarks read slice by slice, and what those scores show per patient.

Materials and Methods: Retrospective secondary analysis of de-identified label files from six imaging benchmarks, including NYU fastMRI, analyzed in 2026. The largest, the 2019 RSNA Intracranial Hemorrhage benchmark, held 752,802 slices from 18,938 patients. A 20-bin relative-position model, locked before any held-out value was computed, was fitted and read at both units on each of 24 independently drawn patient-disjoint holdouts, mean aggregation pre-specified. The primary estimate is that family's mean; one holdout, at a seed fixed in advance, is also reported with a 95% bootstrap confidence interval (CI). The locked protocol was then applied unchanged to a benchmark label file that had not been read.

Results: Over 24 holdouts one score vector read area under the receiver operating characteristic curve (AUC) 0.7381 (0.7350–0.7415) per slice and 0.4551 (0.4424–0.4628) per patient; on the pre-registered holdout, 0.738 (95% CI: 0.733, 0.742) and 0.442 (95% CI: 0.428, 0.458). A constant predictor read exactly 0.500 at both units in every draw. Depth alone read 0.394 per patient; with depth fixed the family read 0.5037 (0.4979–0.5071), covering chance. On the unread benchmark it read 0.7482 per slice against 0.5712 per patient, depth alone 0.5048.

Conclusion: A pixel-blind score vector ranked slices far above chance and patients at or below it, the difference carried by stack depth; the divergence reproduced on a benchmark not previously read, though its mechanism did not.

Introduction

A three-dimensional acquisition is often labeled slice by slice, and performance is then reported by pooling slices into a single area under the receiver operating characteristic curve (AUC). The clinical question is almost always about a patient. The gap is arithmetic: findings are not uniformly distributed along an organ, so a slice-level ranking can be dominated by *where* a slice sits in the stack, and a score vector encoding only that geometry ranks slices well and patients not at all.

None of this is new. Shortcut learning is named [1]: Badgeley et al predicted hip fracture at AUC 0.78 from the radiograph, 0.91 with hospital process features, and 0.52—chance—on a test set balanced across those variables [2]. On the evaluation unit, slice-level cross-validation inflated test accuracy by 29%–55% and reached about 96% on *randomly labeled* data [3, 4]. Stronger baselines have been urged [5].

The central construction used here is prior art. A public 2020 Kaggle notebook on the RSNA-STR Pulmonary Embolism Detection challenge, "Baseline with no image," bins the label rate against relative slice position on the training set and applies it to the test set with no pixels [6]. Yan et al, defining the DeepLesion lesion-type task, publish a "Baseline: Location feature" row at 59.7% eight-class accuracy against their own method's 90.5% [7]. Closest of all, Wallis and Buvat classify tumours in a widely used brain MRI dataset to high accuracy *without information about the tumour*, tracing it to implicit radiologist input in the selection of the 2-D slices [8]—a shortcut carried by which slices were chosen, where the one studied here is carried by where a slice sits among those chosen.

Dataset-level shortcuts are also now catalogued as they are discovered, rather than fixed at a dataset's release [9].

That literature measures the cost of a *wrong*, leaky split rather than what survives a *correct*, patient-disjoint one, and it does not ask what one pixel-blind score vector is worth when the unit of the reading changes. The purpose of this study was to measure one such vector at both units under a patient-disjoint split and to identify the mechanism separating them.

Materials and Methods

Study Design

This was a retrospective secondary analysis of publicly released, de-identified label files from imaging benchmarks released 2016–2024; analyses ran in 2026. Institutional review board approval was not required: no data came from interaction with a living individual and none was identifiable. Age and sex are not distributed with the flagship label file.

Benchmark Eligibility

A dataset entered if the four fields a positional null needs—subject identifier, slice index or z position, per-slice label, train/test assignment—were obtainable without pixel data or a data-use agreement. Of 25 candidates, eight label files across six benchmarks are reported, in two tiers: five *core*, with every field in the benchmark's own pixel-free release, and three *qualified*, with one field absent or reachable only outside it. The flagship is qualified, not core (Table 1).

The six are RSNA 2019 Intracranial Hemorrhage (ICH) [10], fastMRI Prostate, T2-weighted and diffusion-weighted imaging (DWI) arms [11], DeepLesion [12], Duke Breast Cancer MRI [13, 14], PI-CAI [15] and LUNA16 [16]; RSNA ICH is the largest, at 752,802 slices, 21,744 series and 18,938 patients. A seventh, fastMRI+ knee [17], was scored earlier and is withdrawn, its label table being unrebuildable on the original cohort (Table 1).

Every RSNA ICH value rests on a slice ordering and a subject identifier its official release does not contain; both came from a public, license-permissive, pixel-free tabular mirror [18]. Neither was assumed. The official label file was obtained and the mirror reconciled against it row for row: labels agree exactly on every image it holds. That file carries neither of the two fields, so the mirror's groupings and slice order rest on the internal-consistency tests in Table 2, with their power and limits. No image header was read.

Data used in the preparation of this article were obtained from the NYU fastMRI Initiative database (`https://fastmri.med.nyu.edu`) [19, 20]. As such, NYU fastMRI investigators provided data but did not participate in analysis or writing of this report. A listing of NYU fastMRI investigators, subject to updates, can be found at `https://fastmri.med.nyu.edu`. The primary goal of fastMRI is to test

whether machine learning can aid in the reconstruction of medical images.

The Zero-Image Baseline Family

The *primary* baseline was locked before any held-out value was computed: a *positional* model, a 20-bin estimate of the label rate given relative slice position within a volume. Four secondary models were fit on the same training rows: volume size; a depth-limited tree over the remaining non-image columns; a combined position-plus-metadata tree; and each admissible column read singly. A constant prevalence predictor is the chance anchor; columns restating the label, or derived from the images, were excluded. Non-image columns are classified three ways, being not interchangeable: geometry and acquisition, administrative, and legitimate clinical predictors. The primary result rests on the first alone, the only class the flagship label file contains.

Estimator, Aggregation, Intervals, and Nulls

Every baseline produces one score vector per test set, read at *both* units: slice-level AUC, and patient-level AUC after aggregating each subject's scores. The *mean* operator was pre-specified; maximum, top-*k* mean and quantile operators are sensitivity analyses, reported with their distinct-score counts.

Splits are subject-disjoint, using a benchmark's published assignment where one exists. On RSNA ICH, which publishes none, a holdout is 30% of patients with a single fit on the rest and no pooling, so a constant predictor scores exactly 0.500 at both units by construction. The *primary estimate is the mean over 24 independently drawn such holdouts*, each with its own fit, and its uncertainty statement is the range over that family, which conditions on no one split or model. One draw, at a seed fixed before any held-out value was computed, is also reported alone with a 95% percentile bootstrap CI resampling *patients* over 2000 replicates; it is a reproducible reference point, not a tighter estimate (Table 3).

The pooled out-of-fold estimator used in an earlier round is used nowhere here: each fold emits its own training prevalence, so fold identity becomes rankable and its constant predictor sits at 0.492 per slice. Each baseline is read against its own permutation null, which for the positional model shuffles labels *within each series*, preserving prevalence, subject clustering and stack depth.

A Secondary Comparison Statistic

As a *secondary* description only, the share of a published margin over chance reached by the locked baseline—(baseline − chance) / (published − chance), against each benchmark's strongest published system—is reported in the supplement (Fig S1). It is not inferential and no average is taken: those margins sit on different splits and different metrics against different chance anchors. No claim here rests on it.

Statistical Analysis

This is an estimation study: no significance test was pre-specified and no *P* values are reported. Analyses used Python 3.14 (Python Software Foundation, Wilmington, Del), NumPy 2.4, pandas 3.0 and, for the tree baselines, scikit-learn 1.8.

Results

One Locked Score Vector, Read at Two Units

No published number enters this section. On RSNA ICH the locked 20-bin positional model was fitted and read on each of 24 independently drawn patient-disjoint holdouts, each 30% of patients with a single fit on the remaining 13,257. Read at the slice level the resulting vectors average AUC 0.7381 (range, 0.7350 to 0.7415); read at the patient level, after the pre-specified mean aggregation, the same vectors average 0.4551 (range, 0.4424 to 0.4628). Nothing changes between the readings except the unit. On the pre-registered holdout of 5,681 patients (226,663 slices, 32,258 positive; 2,306 positive patients) the same pair reads 0.738 (95% CI: 0.733, 0.742) and 0.442 (95% CI: 0.428, 0.458); the difference there is 0.295, and over the six official labels the gaps run 0.181 to 0.318 (Table 2, Fig 1A). A constant predictor reaches exactly 0.500 at both units, on all six labels and in all 24 holdouts.

At the patient unit the six labels do not all behave alike. Three intervals lie wholly below chance (any, subarachnoid, subdural) and three cross it; only epidural's point estimate is above chance, 0.528 (95% CI: 0.472, 0.579) on 108 positive patients. The within-series permutation null is 0.505 per slice, so the slice-level excess is 0.233; per patient that null is 0.580, *above* chance, so the observed 0.442 sits 0.138

below what this estimator reaches with nothing to find.

The Mechanism Is Stack Depth

A patient's mean is one fitted 20-value curve sampled at that patient's slice positions, so it is largely a function of how many slices the stack holds; and depth tracks the label here, reading 0.394 (95% CI: 0.380, 0.409) per patient on its own—the direction the mean inherits. Held at fixed depth the anti-ranking goes. Over the family that reading is 0.5037 (SD 0.0021; range, 0.4979 to 0.5071), which *covers* chance; on the pre-registered draw it is 0.5071 (95% CI: 0.5013, 0.5129) within exact stack depth over 29 informative strata, and 0.5024 (95% CI: 0.4867, 0.5178) within 5-slice strata (Table 2, Fig 1B). That draw's interval excludes 0.500 and is still not read as chance; what it does is sit at its own null, the within-series permutation null at fixed depth being 0.5055, 0.0016 below it. Per label the depth-conditioned reading runs 0.4999 to 0.5108. Depth alone averages 0.4031 over the family, and the constant predictor 0.5000, SD 0.0000 (Table 3, Fig 1C).

Which Uncertainty Statement Applies

Neither uncertainty statement replaces the other: at the patient unit the bootstrap CI on the pre-registered draw is the wider (0.030 against the family range of 0.020) and the narrower on two of the six labels. That draw sits at the edge of its own family—lowest of 24 on the patient reading, highest on the depth-conditioned one—because the depth confound is unusually strong in it. Its seed was fixed before any held-out value was computed, so reporting the family mean as primary is what keeps that from mattering (Table 3, Fig 1C).

Aggregation, and the Rest of the Baseline Family

Six alternatives to the mean were run (Table 3). Maximum and top-1 mean are degenerate under a single fit and read exactly 0.500. Three read above chance—top-3 mean 0.514, top-5 mean 0.513 and the 90th percentile 0.556—so "at or below chance" holds for the mean and the 75th percentile (0.480) and not for those three, which take 4, 9 and 13 distinct values and are near-degenerate readings of stack depth. The strongest secondary model is volume size, at 0.601; every column here is a geometry or acquisition field, which is why the primary result is about geometry alone.

The Same Reading on Every Other Benchmark-Arm

Every arm reported here is scored by a single fit on a subject-disjoint split, and the constant predictor is exactly 0.500 on all of them. On its own published split, fastMRI Prostate reads slice 0.854 against patient 0.506 on T2 and 0.851 against 0.424 on DWI. All eight DeepLesion body-part arms rank above chance at both units (slice 0.691–0.977, patient 0.642–0.954) and none diverges, which is expected: the label there is the anatomic region, so relative position is legitimate task information. LUNA16 moves little at either unit and its slice-level interval reaches chance (0.531; 95% CI: 0.486, 0.573); Duke Breast reads 0.821 (95% CI: 0.799, 0.841) per slice, its patient-level value undefined, all 922 of 922 patients being positive (Table 4, Fig 2).

The Same Divergence on a Benchmark Not Previously Read

The locked protocol was then applied, nothing re-tuned, to a label file that had not been opened: PI-CAI's 1,295 human-expert lesion delineation volumes, yielding 28,648 slices over 1,273 patients. An earlier round recorded it as not estimable, true of the public marksheet and false of the benchmark. Over 24 holdouts of the identical design the pixel-blind vector reads 0.7482 (range, 0.6974 to 0.7744) per slice against 0.5712 (range, 0.4995 to 0.6215) per patient, a gap of 0.177, the constant predictor exactly 0.500 in every draw (Table 4, Fig 2). The divergence reproduces on a different organ, modality and institution set. The mechanism does not: depth alone reads 0.5048 here, and the depth-conditioned reading is degenerate at exactly 0.500 over 17 strata, as on fastMRI Prostate, because within a depth stratum every patient's mean is one number.

Sensitivity to the Two Analysis Choices

The two units do not respond alike to the choices the baseline requires. Across 5 to 50 position bins the slice-level reading spans 0.028; crossed with six aggregation operators, the patient-level reading over the same fits spans 0.415 to 0.652, a range of 0.237 (Fig 3). At 50 bins every operator reads above chance, a bin then holding about one slice so that the aggregate tracks depth directly. Twenty bins with mean aggregation were pre-specified, and every patient-level statement here concerns that cell; the slice-level statement does not.

The share of a published margin over chance the locked baseline reaches, on every benchmark with a peer-reviewed comparator, is reported in the supplement (Fig S1).

Discussion

The unit at which a three-dimensional benchmark is read decides what its number means. Over 24 patient-disjoint holdouts a single pixel-blind score vector read 0.7381 per slice and 0.4551 per patient; only the first is what a paper would print. The mechanism is identified rather than inferred: mean aggregation carries stack depth into a per-patient score, depth is associated with the label here, and with depth held fixed the same vector reads 0.5037 across the family, covering chance. Read the patient-level value as a null, not as a reversal.

Three protocol choices carry that claim. Every holdout is a single fit, under which a constant predictor is exactly 0.500 by construction; the pooled out-of-fold alternative put that floor at 0.492, a ranking artifact of pooling rather than a measurement. The baseline was locked before the held-out data were touched, so nothing is selected on the outcome. And the primary estimate is the mean over the family rather than any one draw, because a single draw—including the one whose seed was fixed in advance—can sit at the edge of what the procedure produces. The cost is precision: intervals widen about 1.7 to 1.9 times.

What survives the analysis choices is the slice-level reading, not the patient-level one. That asymmetry is itself the practical result: a slice-pooled AUC on this benchmark is reachable without pixels under every binning tried, whereas how far the per-patient reading falls depends on the aggregation operator. Where a pixel-blind model ranks well and does *not* diverge between units, the right reading may be that the task is legitimately positional, as DeepLesion's is.

Applying the locked protocol unchanged to a label file that had not been read separates two claims that one benchmark cannot. The divergence generalised—0.75 per slice against 0.57 per patient on prostate MRI, a different organ, modality and institution set from head CT. The depth mechanism did not: depth alone reads 0.5048 there. The finding to carry forward is that a slice-pooled and a per-patient AUC of one pixel-blind vector can differ widely on unrelated benchmarks, while what produces that difference must be established per benchmark rather than assumed.

Badgeley et al's control is *negative*—balance the confounders and watch the model collapse to 0.52 [2]. This one is *positive*: fit a model on the confounder alone and see what it reaches. Against the leakage literature the distinction is the split, theirs wrong and this one patient-disjoint [3, 4]; against Yan et al it is the source of the position, theirs from a body-part regressor on the image and this one from a published column [7].

For a radiologist reading an imaging AI paper, the operative question is what a reported AUC was computed over. On the hemorrhage benchmark the slice-pooled number stood 0.181 to 0.318 above the per-patient number from the same scores, and the per-patient number is the one aligned with a per-patient decision—a statement about these benchmarks, since no regulatory filing was examined. For benchmark publishers, three fields already in the released label file—a subject identifier, a slice index or z position, and the train/test assignment—make every one of these checks runnable by anyone. What this protocol cannot do is detect shortcuts living in the pixels.

Limitations. The construction is prior art [6, 7]. The patient-level reading is not robust to either analysis choice the baseline requires, unlike the slice-level one, and every patient-level statement here is a statement about the pre-specified cell (Fig 3). The depth mechanism is established on the flagship alone; on the two benchmarks where depth could be conditioned on it was degenerate, so the mechanism is not shown to be general. The flagship's subject identifier and slice index are not in its own release but in a public tabular mirror. Its labels reconcile exactly against the official release, but that file carries neither field, so the mirror's groupings and slice order still rest on the internal tests in Table 2. A seventh benchmark scored in an earlier round, fastMRI+ knee, is withdrawn because it cannot be reproduced on its original cohort. LUNA16's candidate list is detector-produced, so "zero image" there means "given the published candidate list." The PI-CAI comparison is across cohorts, and no peer-reviewed comparison is matched. The candidate universe has a stated entry criterion and a published exclusion list but is not externally registered, and two included benchmarks already carried results when it was written (Table 1). Age and sex are absent from the flagship label file, and no subgroup analysis is reported.

In summary, one pixel-blind score vector ranked slices well above chance and patients at or below it, the difference carried by stack depth rather than by any reversal of the positional signal. Three fields a benchmark already publishes make that checkable by anyone, without the images.

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Acknowledgments are withheld from this version because they would identify the authors, and will be supplied on acceptance.

Use of large language models. Large language model assistance was used in two ways: in drafting and editing this manuscript, and in writing the analysis code that produces every number in the Results. The tool was **Claude (Opus 5), Anthropic**, accessed 2026-07-27 to 2026-08-14. No value reported here was generated by a language model: every number in the Results is a deterministic output of the released code, run on a published label file, and the flagship estimate was reproduced by a second implementation sharing no code with the primary one, which ran the same pre-specified protocol on its own holdouts and returned a family mean agreeing to 0.0003 AUC per slice and 0.005 per patient, with a constant predictor of exactly 0.500 in every draw of both families (Table 3). Readers are directed to the released code rather than asked to take the provenance on trust.

Data and code availability. The audit tool, the per-benchmark output payloads, the label-table preparation scripts, the second implementation of the intracranial hemorrhage unit-collapse computation, and the assembler that builds the cross-study comparison are released under a Creative Commons Attribution licence in a public repository; the repository name, its address, and the commit identifier for the revision used are withheld from this version for anonymized review and will be stated in the Materials and Methods on acceptance, and are available to the editors on request. No benchmark's pixel data are redistributed. The worked-example label files from the NYU fastMRI Initiative are available by application to that initiative under its data sharing agreement, which forbids onward distribution of the dataset or of files derived from it; all other label files used here are public.

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Figure Legends

Figure 1. One locked pixel-blind score vector on the RSNA 2019 Intracranial Hemorrhage benchmark, read on a frozen patient-disjoint holdout (13,257 training patients; 5,681 held-out patients, 226,663 held-out slices; single fit, no pooling). **(A)** Slice-level area under the receiver operating characteristic curve (AUC) joined to patient-level AUC after mean aggregation for each of the six official labels, with 95% patient-clustered bootstrap intervals (2000 replicates) and the within-series label permutation null at the patient unit. Nothing changes between the two readings of a pair except the unit at which the ranking is performed. Three patient-level intervals lie wholly below chance and the other three cross it; epidural, on 108 positive patients, is the only point estimate above chance. The patient-level null lies above chance because permuting within a series preserves stack depth. **(B)** The mechanism, for the "any hemorrhage" label: the same patient-level score read unstratified, within exact stack depth, and within 5-slice depth strata, with stack depth read alone for comparison. Holding depth fixed removes the sub-chance reading; the depth-conditioned value of 0.5071 (95% CI: 0.5013, 0.5129) excludes 0.500 but sits 0.0016 above the within-series permutation null computed at fixed depth, 0.5055 (range, 0.5035 to 0.5068), shown as the shaded band, which its own interval spans. **(C)** The same four quantities over the 24 independently drawn holdouts of the identical design that furnish the primary estimate. Points are the individual holdouts, the bar is the family range, and the open marker is the pre-registered holdout of panels A and B, which sits at the edge of the family on the patient-level and depth-conditioned readings; that is why the family mean and not this draw is reported as primary. The constant predictor is exactly 0.500 in every holdout, which is the property the superseded pooled out-of-fold estimator did not have.

Figure 2. The same pixel-blind score vector read at two units, for every audited benchmark-arm. Each point is one arm: its slice-level area under the receiver operating characteristic curve (AUC) against the patient-level AUC of the identical score vector after mean aggregation, with 95% subject-clustered bootstrap intervals on both axes (replicate counts in Table 4). The dashed diagonal is the line on which the two units agree, and the grey stem below each point is the distance that arm falls from it. All eight DeepLesion body-part arms are plotted, not one of them: they are the arms on which no divergence occurs, and that is expected rather than anomalous, because the label there is the anatomic region and relative slice position predicts it at either unit. Values are for the locked 20-bin positional baseline except PI-CAI, whose

positional baseline is exactly 0.500 on the public marksheet, which carries no slice index, and whose secondary tree over that file's four non-image columns—two of them clinical, age and prostate-specific antigen level—is drawn instead. That benchmark's slice-level arm, read from its 1,295 lesion delineation volumes under the same locked protocol, is plotted separately and at both units. Every arm plotted is scored by a single fit on a subject-disjoint split and its constant predictor is exactly 0.500. The two arms defined at only one unit are drawn in the margins rather than in the field, so that a value that does not exist cannot be read as a low one: PI-CAI's label file carries no slice index, and Duke Breast's patient-level AUC is undefined because all 922 of its 922 patients are positive.

Figure 3. How the two units respond to the two analysis choices the positional baseline requires. **(A)** Slice-level AUC against the number of position bins: 0.717, 0.733, 0.738, 0.738 and 0.745 over 5, 10, 20, 30 and 50 bins, a range of 0.028. **(B)** Patient-level AUC over the same bin counts crossed with six aggregation operators, on the same frozen holdout of 5,681 patients and the same single fit. The boxed cell, 20 bins with mean aggregation, is the pre-specified analysis reported throughout; every other cell is a sensitivity analysis and none was used to choose it. Colour is centred on 0.500. The patient reading spans 0.415 to 0.652, a range of 0.237, against 0.028 at the slice unit for the identical fits. Two features carry the paper's caveat rather than its claim: at 50 bins every operator reads above chance, because a bin then holds about one slice and the aggregate begins to track stack depth directly; and the operators that read near exactly 0.500 are degenerate, taking only a handful of distinct values across all 5,681 patients. The slice-level reading is nearly invariant to both choices. The patient-level reading is not, and any patient-level statement made here is a statement about mean aggregation at 20 bins.

Table 1: Benchmark Eligibility—the Candidate Universe, the Four Required Fields, and Every Disposition

Candidate	S	Z	L	P	Disposition and reason
Included, tier 1—*core*: all four fields in the benchmark's own pixel-free release					
fastMRI Prostate, T2-weighted	Y	Y	Y	Y	Included. In-file split, subject-disjoint
fastMRI Prostate, DWI	Y	Y	Y	Y	Included
DeepLesion	Y	Y	Y	Y	Included, 8 body-part arms and 1 eight-class arm. Publishes normalized lesion z directly
LUNA16	Y	Y	Y	Y	Included, 1 arm. Candidate list is detector-produced, so "zero image" means "given the published candidate list"
PI-CAI, slice level	Y	Y	Y	Y	Included, 1 arm. The four fields come from the benchmark's own pixel-free release once the 1,295 human-expert lesion delineation volumes are read; those are integer label masks, and no image intensity is opened
Included, tier 2—*qualified*: one field absent, or reachable only outside that release					
RSNA 2019 Intracranial Hemorrhage	Y*	Y*	Y	Y	**The flagship, and qualified rather than core.** Official release carries the label only; subject identifier and slice index come from a public, license-permissive, pixel-free tabular mirror. Eligible under "obtainable," not under "in the benchmark's own release"
Duke Breast Cancer MRI	Y	Y	Y	N	Included, 1 arm, on the data owners' own published slice definition. All subjects positive, so patient-level AUC is undefined and is reported as undefined
PI-CAI, case level	Y	N	Y	Y	Included, 1 arm, on the public marksheet, which carries no slice index. The locked positional baseline is not estimable *on this file*; it is estimable on the same benchmark's delineation volumes, which are the separate tier-1 arm above
Scored in an earlier round, withdrawn here—reproducibility, not eligibility					
fastMRI+ knee	Y	Y*	Y	N	**Withdrawn, 2 arms.** Eligible, and scored in an earlier round under the superseded pooled estimator. Its prepared label table was lost and only 155 of the 199 roster volumes remain locally, so the arm cannot be rebuilt on its original cohort and re-scored under this submission's estimator. Withdrawn rather than carried on the superseded estimator. The two arms it contributed read 0.873 and 0.801 per slice
Excluded on eligibility—criteria recorded before that run's results					
CQ500	Y	N	N	N	L fails: three radiologist reads per scan, no per-slice label
BraTS, KiTS, Medical Segmentation Decathlon, AMOS, TotalSegmentator	Y	Y*	Y*	Y	Pixel-free download fails (masks ship with the images) and the benchmark metric is a spatial overlap score, so no slice-level number exists to compare against
PROSTATEx	Y	N	Y	Y	Z fails: finding position is in patient coordinates in millimeters, needing image headers to convert
Prostate158	Y	Y*	Y*	N	L fails and power fails: no slice-level classification task exists
MRNet	Y	N	N	Y	Z and L fail by construction
Eligible or near-eligible, not reached—access or power, not eligibility					
RSNA 2023 Abdominal Trauma	Y	Y	Y	Y	All four fields present; blocked by a click-through agreement that was not accepted. The strongest untaken target
RSNA 2022 Cervical Spine	Y	Y	Y	Y	Per-slice labels for 235 of 3,112 studies only; underpowered, plus the same access blocker
fastMRI+ brain	Y	Y*	Y	N	73 of 1,001 roster volumes held; underpowered
Added to the universe after other results were seen, then excluded on a measurement					
RSNA-STR Pulmonary Embolism	Y	**N**	Y	Y	Label file obtained and verified genuine against a published row count, then excluded: Z is absent, and neither the file's row order (run-length ratio 0.974) nor its identifier order (1.001) carries positional information. Peer-reviewed numbers on it are exam-level on a private test set whose labels were never released
Named but unassessed—no label file obtained; neither included nor excluded					
Six further candidates	—	—	—	—	Recorded in the candidate list as unverified; no field inventory exists

Note.—**Tier 1 files satisfy the entry criterion as written; tier 2 files do not, and each is carried for a stated purpose with its shortfall named.**

The flagship sits in tier 2 because its subject identifier and slice index are not in its own release. That placement does not weaken the primary result, which compares two readings of one score vector *within* the benchmark and so requires the mirror to be internally consistent rather than canonical; Table 2 tests that consistency and states what those tests cannot establish. It does constrain the cross-study comparison, where a tier-2 file is being set beside a number computed by others on the release itself. S = subject identifier; Z = slice index or z position; L = per-slice label; P = train/test assignment. Y = present in a pixel-free public file; Y* = obtainable, but not from the benchmark's own release or not without an image file header read; N = not obtainable. AUC = area under the receiver operating characteristic curve; DWI = diffusion-weighted imaging.

Twenty-five datasets were named, counting the five segmentation benchmarks individually; 19 were assessed against the four fields and 6 were not. The PI-CAI slice-level arm was recorded in an earlier round as eligible but not reached, on the ground that its label volumes needed a reader that was unavailable; that was a tooling limitation, not an eligibility one, and it has since been read under the same locked protocol with nothing re-tuned for it. Every exclusion is a named field failure; none followed a computed baseline, and no candidate was dropped after its baseline had been computed. The candidate list, its entry criterion and its per-candidate exclusion reasons were recorded before 7 of the 8 included label files were scored; that ordering rests on file timestamps rather than an external registration, and two included benchmarks—fastMRI Prostate and DeepLesion— already carried results when the list was written. The audit is therefore reported as having a stated entry criterion and a published exclusion list, not as pre-specified. The one case added to the universe after other results were seen was then excluded, which runs against rather than with the interest of this study.

Table 2: RSNA 2019 Intracranial Hemorrhage—One Locked Pixel-Blind Score Vector Read at Two Units on a Frozen Patient-Disjoint Holdout, All Six Official Labels

Label	Slice AUC	Patient AUC (mean)	Gap	Patient AUC, depth fixed
Any hemorrhage	0.738 (0.733, 0.742)	0.442 (0.428, 0.458)	0.295	0.5071 (0.5013, 0.5129)
Epidural	0.709 (0.687, 0.729)	0.528 (0.472, 0.579)	0.181	0.5044 (0.4819, 0.5306)
Intraparenchymal	0.753 (0.746, 0.760)	0.483 (0.467, 0.501)	0.270	0.4999 (0.4935, 0.5061)
Intraventricular	0.803 (0.798, 0.809)	0.485 (0.467, 0.504)	0.318	0.5062 (0.4992, 0.5135)
Subarachnoid	0.695 (0.688, 0.703)	0.467 (0.449, 0.485)	0.228	0.5091 (0.5016, 0.5169)
Subdural	0.720 (0.715, 0.725)	0.448 (0.429, 0.467)	0.272	0.5108 (0.5020, 0.5195)

Any hemorrhage, patient level, with stack depth held fixed

Patient-level reading	AUC	Pairs compared
Mean aggregation, unstratified	0.442 (0.428, 0.458)	7,782,750
within exact stack depth (29 strata)	0.5071 (0.5013, 0.5129)	901,972
within 5-slice depth strata	0.5024 (0.4867, 0.5178)	2,253,183
Stack depth alone	0.394 (0.380, 0.409)	7,782,750
Maximum aggregation	0.500 exactly	7,782,750

Note.—AUC = area under the receiver operating characteristic curve; CI = confidence interval. The benchmark holds 752,802 slices, 21,744 series and 18,938 patients; this table is computed on one frozen patient-disjoint holdout, 30% of patients drawn at a pre-set seed, with a single fit on the remaining 13,257 patients and no pooling. The analysis code is released, so every value here is reproducible from the released label table; the seed was set before any held-out value was computed, but that ordering rests on working-file timestamps rather than an external registration, and is reported on the same footing as the eligibility ordering in Table 1. The holdout carries 5,681 patients and 226,663 slices. Slice-level positive counts are 32,258 (any), 1,163 (epidural), 10,915 (intraparenchymal), 7,901 (intraventricular), 10,991 (subarachnoid) and 13,701 (subdural); patient-level positive counts are 2,306, 108, 1,408, 1,003, 1,032 and 987 of 5,681. Data in parentheses are 95% percentile intervals from 2000 bootstrap replicates resampling patients, in both blocks; the across-holdout spread, a different quantity that is wider at the patient unit on two of the six labels and narrower on the other four, is in Table 3. Nothing changes between the slice and patient columns except the unit at which the ranking is performed. *The constant predictor scores exactly 0.500 at both units, for all six labels and in all 24 holdouts.* Under a single fit that is true by construction, and it is the reason this estimator replaced the pooled out-of-fold estimator used in an earlier round, whose constant predictor scored 0.492 per slice because each fold emitted its own training prevalence and fold identity became rankable on its own. The within-series label permutation null is 0.505 at the slice level and 0.580 at the patient level; the patient-level null lies *above* chance because permuting within a series preserves stack depth, so the depth confound survives it. At fixed depth that null is 0.5055 (range over 20 permutations, 0.5035 to 0.5068). The observed 0.5071 sits 0.0016 above that mean and 0.0003 above the top of that range, and its own interval, 0.5013 to 0.5129, spans the whole of it, so the depth-conditioned reading is reported as sitting at its own null rather than at chance. The depth-conditioned column is printed to four decimals for that reason. Depth-fixed values are pooled within-stratum concordance with ties counted as half, over the pairs shown; stack depth is a patient's slices per volume. Maximum aggregation is exactly degenerate under a single fit: every volume holds at least 20 slices and spans all 20 position bins, so all 5,681 patients take one score. Three further checks ran on this same holdout. The slice-level reading is not a binning artifact: 0.717, 0.733, 0.738 and 0.745 over 5, 10, 20 and 50 bins, with a fit-free centrality score, which uses no training data at all, at 0.735. The model is not itself overfitting: apparent AUC on the training rows 0.738 against held-out 0.738. And metadata adds nothing here: the combined position-plus-metadata tree reaches 0.719 per slice against the locked model's 0.738. *Provenance of the slice ordering and the subject identifier*, neither of which the official release contains. Ordering, over the 8,377 series that carry three or more positive slices, which are the series on which a run-length test has any power: positive slices form 1.384 runs per series against 7.167 expected under random placement, a ratio of 0.193 against 1.001 (SD 0.002) over 20 randomized orderings, $z = -452$; the same test fires on all five subtypes at ratios 0.175 to 0.246, the rarest (epidural, 331 series) at $z = -86$; 99.5% of those series are individually more contiguous than chance and 74.8% form a single run. Power: relocating 0%, 5%, 10%, 20%, 30% and 100% of slices at random within series gives ratios 0.193, 0.233, 0.301, 0.444, 0.576 and 1.001, so the pre-set abort threshold of 0.50 is crossed near 24% and certifies only against wholesale mis-ordering, while as a continuous statistic the test moves about 20 SDs at 5% corruption. Orientation is coherent across series: mean relative position of positive slices 0.5239 against 0.4998 (SD 0.0011) when a random half of series are reversed, $z = +22$. Subject identifier, on 1,758 subjects holding 4,564 series, where series and studies are 1:1 so every pair spans two examinations: agreement on reconstruction intercept 0.977 against 0.515 (SD 0.007) with subject labels permuted, $z = 68$; on hemorrhage present 0.801 against 0.500, $z = 44$, holding at 0.800 against 0.504 among the 5,023 pairs already matched on intercept, $z = 25$; on series depth 0.188 against 0.098, $z = 40$. Structure: 21,743 of 21,744 series have a gapless within-series counter and the single gapped series implies exactly one missing row, reconciling the 752,802 rows held against the 752,803 the comparator describes. *Reconciliation against the official release.* The official label file was subsequently obtained and the mirror checked against it row for row. It carries 4,516,842 rows, of which 24 are exact duplicates over 24 identifiers with no conflicting labels; deduplicated it describes 752,803 images. The mirror holds 752,802 of them and contains no image the official file does not. Across all six official labels on every shared image—4,516,812 label comparisons—there is *no* disagreement, and the per-label positive counts match exactly: 3,145 epidural, 36,118 intraparenchymal, 26,205 intraventricular, 35,675 subarachnoid, 47,166 subdural and 107,933 any. The single image the mirror lacks is the one the gapless-counter test above predicted before the official file was seen:

that test located exactly one gapped series, missing exactly one index, and the official release contains exactly one image absent from the mirror. What this establishes is that the mirror reproduces the official label of every image it claims to hold. What it cannot establish is the mirror's series and patient groupings or its within-series slice order, because the official file carries neither a subject identifier nor a slice index—which is why the mirror is used at all. Those still rest on the run-length and orientation tests above. No image header was read at any point.

Table 3: Estimator, Aggregation and Baseline Sensitivity on the Intracranial Hemorrhage Benchmark (“Any Hemorrhage”)

A. Pipeline uncertainty: the whole procedure repeated on independent holdouts

Family	Draws	Slice AUC	Patient AUC (mean)	Patient, depth fixed	Constant
Primary implementation	24	0.7381 [0.7350, 0.7415]	0.4551 [0.4424, 0.4628]	0.5037 [0.4979, 0.5071]	0.5000
Second implementation	8	0.7378 [0.7359, 0.7401]	0.4603 [0.4383, 0.4741]	0.5005 [0.4955, 0.5050]	0.5000
Frozen holdout, bootstrap	1	0.7377 (0.7333, 0.7417)	0.4424 (0.4281, 0.4580)	0.5071 (0.5013, 0.5129)	0.5000

B. Aggregation operators, frozen holdout, 5,681 **patients**

Operator	Patient AUC (95% CI)	Distinct patient scores
Mean (pre-specified)	0.442 (0.428, 0.457)	275
Maximum	0.500 exactly	1
Top-1 mean	0.500 exactly	1
Top-3 mean	0.514 (0.501, 0.528)	4
Top-5 mean	0.513 (0.499, 0.526)	9
75th percentile	0.480 (0.467, 0.493)	6
90th percentile	0.556 (0.541, 0.571)	13

C. The four secondary zero-image models, frozen holdout

Model	Slice AUC	Patient AUC (mean)
Locked positional, 20 bins (primary)	0.738	0.442
Volume size	0.583	0.601
Metadata tree	0.521	0.527
Position and metadata combined	0.719	0.573
Single column: imaging plane	0.500	0.500
Single column: reconstruction slope	0.500	0.500
Single column: reconstruction intercept	0.536	0.553

Note.—AUC = area under the receiver operating characteristic curve; CI = confidence interval; SD = standard deviation. *Block A.* Values in square brackets are the minimum and maximum over the independent holdouts of that family, not confidence intervals: each holdout is a fresh draw of 30% of patients with its own fit, so the range describes pipeline uncertainty and does not condition on any one fitted model. Values in round parentheses on the last line are a 95% percentile bootstrap CI over 2000 replicates resampling patients, which does condition on the one fit and the one holdout. The two are not ordered in a fixed direction: on “any” the bootstrap CI is the wider (0.0299 against an across-holdout range of 0.0203), on epidural and subdural the across-holdout range is the wider, and both are reported for every primary value. Under the superseded pooled estimator the patient-level interval was 0.0159 wide and did understate holdout-to-holdout movement; the frozen estimator’s wider interval is what repairs that. SDs over the 24 draws are 0.0016 (slice), 0.0059 (patient), 0.0021 (depth fixed) and 0.0000 (constant). The two families ran the same pre-specified protocol in two implementations sharing no code, on their own draws: their means agree to 0.0003 per slice and 0.005 per patient, and the constant predictor is exactly 0.500 in every draw of both. Under the superseded pooled out-of-fold estimator that constant was 0.492, so this is also the check that the artifact is gone rather than reduced. The pre-registered holdout is the lowest of its 24 on the patient-level reading and the highest on the depth-conditioned reading; its depth-alone value is second lowest of 24, and depth alone correlates 0.72 with the patient-level reading across draws. Its seed was fixed before any held-out value was computed, so that position is a property of the draw rather than a choice among draws; it is nonetheless the reason the family mean, and not this draw, is reported as the primary estimate. *Block B.* The number of distinct patient scores is printed for every operator because three of them are near-degenerate: top-3 mean, top-5 mean and the 90th percentile take 4, 9 and 13 values across 5,681 patients, so their above-chance readings are readings of stack depth and not independent evidence. Maximum and top-1 mean take a single value and are exactly 0.500. The mean row is the same point estimate as block A’s frozen holdout; its interval differs in the third decimal (0.457 against 0.458) because this block was bootstrapped in its own run under its own seed, and both are reported as computed rather than reconciled. *Block C.* The positional model was locked as primary before any held-out value was computed; the other rows are secondary and are reported for completeness. On this benchmark the locked model is also the strongest of the family on every one of the six labels, so the lock costs nothing here. Every column in this label file is a geometry or acquisition variable; it carries no administrative and no clinical column, which is why the primary result is a statement about geometry alone.

Table 4: Slice Level versus Patient Level for the Same Pixel-Blind Score Vector, Every Audited Benchmark-Arm, with the Estimator That Produced Each

Benchmark-arm	Estimator	Slice AUC	Patient AUC (mean)	Constant	Class
RSNA ICH, any hemorrhage	Frozen holdout	0.738 (0.733, 0.742)	0.442 (0.428, 0.458)	0.500	G
fastMRI Prostate, T2-weighted	Published split	0.854 (0.811, 0.891)	0.506 (0.382, 0.629)	0.500	G, A
fastMRI Prostate, DWI	Published split	0.851 (0.814, 0.888)	0.424 (0.293, 0.551)	0.500	G, A
DeepLesion, pelvis vs rest	Published split	0.977 (0.969, 0.984)	0.954 (0.939, 0.967)	0.500	G, C
mediastinum vs rest	Published split	0.890 (0.871, 0.906)	0.814 (0.782, 0.844)	0.500	G, C
abdomen vs rest	Published split	0.886 (0.868, 0.903)	0.807 (0.774, 0.841)	0.500	G, C
kidney vs rest	Published split	0.877 (0.859, 0.894)	0.838 (0.802, 0.870)	0.500	G, C
liver vs rest	Published split	0.876 (0.860, 0.890)	0.777 (0.741, 0.812)	0.500	G, C
lung vs rest	Published split	0.872 (0.852, 0.892)	0.803 (0.768, 0.836)	0.500	G, C
soft tissue vs rest	Published split	0.831 (0.777, 0.878)	0.811 (0.769, 0.850)	0.500	G, C
bone vs rest	Published split	0.691 (0.618, 0.763)	0.642 (0.555, 0.726)	0.500	G, C
Duke Breast, owner-defined slice task	Frozen holdout	0.821 (0.799, 0.841)	Undefined	0.500	G, A
LUNA16 candidates	Frozen holdout	0.531 (0.486, 0.573)	0.553 (0.479, 0.628)	0.500	G
PI-CAI, slice level	Frozen holdout	0.773 (0.747, 0.796)	0.598 (0.530, 0.667)	0.500	G
PI-CAI, case level	Published split	Not applicable	0.500 exactly (locked); 0.692 (0.626, 0.755) secondary	0.500	G, A, C

Note.—AUC = area under the receiver operating characteristic curve; DWI = diffusion-weighted imaging; ICH = intracranial hemorrhage. Data in parentheses are 95% percentile bootstrap intervals resampling subjects, from 2000 replicates. Values are for the locked 20-bin positional baseline throughout. "Undefined" on Duke Breast means the patient-level AUC does not exist because all 922 of its 922 patients are positive; "not applicable" on PI-CAI means that label file carries no slice index, so no slice-level reading exists. *Estimator.* "Frozen holdout" is one subject-disjoint holdout with a single fit and no pooling; "published split" is the benchmark's own train/test assignment, also a single fit. *Every row in this table is one or the other, and the constant predictor is exactly 0.500 on every one of them.* An earlier revision carried four arms on a superseded pooled out-of-fold estimator whose constant predictor was 0.455 to 0.483 rather than 0.500. Two of those four, Duke Breast and LUNA16, are reported here re-scored: their prepared label tables had been lost, and were rebuilt from the public source releases and verified byte-identical against the SHA-256 recorded by the original run before being re-scored under the protocol above. The other two, both fastMRI+ knee arms, could not be rebuilt on their original cohort—155 of the 199 roster volumes remain locally—and are withdrawn rather than carried defective (Table 1). Re-scoring moved Duke Breast from 0.823 to 0.821 and LUNA16 from 0.534 to 0.531 at the slice unit; LUNA16's interval now reaches chance, which the pooled interval did not. *Class* records which kinds of non-image variable that label file makes available to the zero-image family: G = geometry and acquisition; A = administrative; C = legitimate clinical non-image predictor. The flagship arm is G only, which is what makes the primary result a statement about geometry alone. The clearest administrative result is the fastMRI Prostate T2 arm, where the column recording which distribution package a subject shipped in ranks patients at 0.596. *DeepLesion.* All eight body-part arms are listed and all eight are carried into Fig 2; no arm is singled out. Every one lies above chance at both units, which is expected rather than anomalous, because the label is the anatomic region and relative position is legitimate task information for it. The eight share one split, one score construction and the same 665 patients, and are not independent of one another. That benchmark also supplies age and sex, which is why its class includes C. *PI-CAI.* The locked positional baseline is exactly 0.500 because that label file carries no slice index. The secondary value is a depth-3 tree over its four non-image columns—patient age, prostate-specific antigen level, center and scan year—and the fitted tree splits on all four. Read one column at a time on this split, patient-level AUC is 0.636 for age, 0.638 for prostate-specific antigen level, 0.552 for scan year and 0.547 for center. The two clinical columns carry the row, so it is a clinical-variable baseline; the 0.692 cannot be partitioned between the classes from the retained output, and no acquisition-confounding inference is drawn from it anywhere.

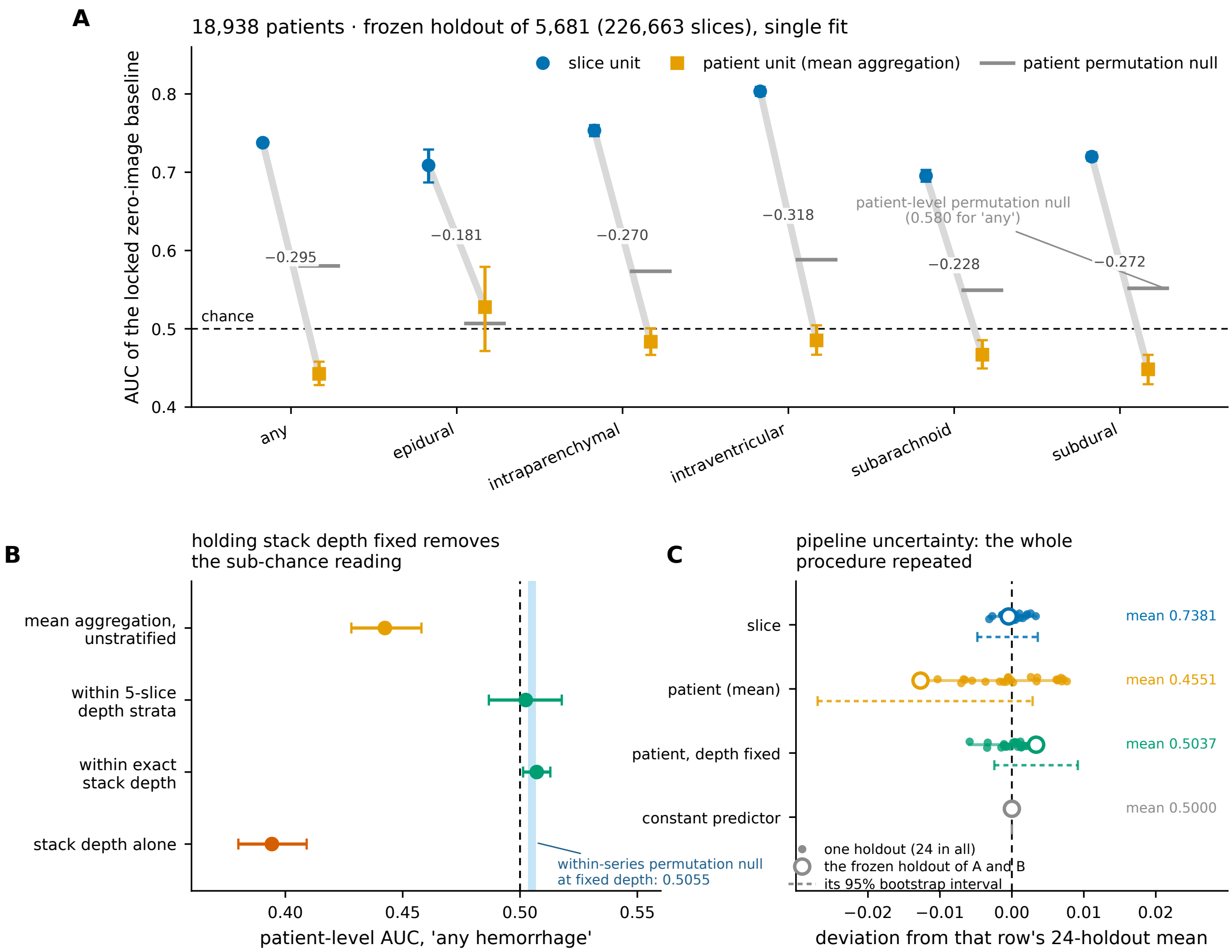

Figure 1. One locked pixel-blind score vector on the RSNA 2019 Intracranial Hemorrhage benchmark, read on a frozen patient-disjoint holdout (13,257 training patients; 5,681 held-out patients, 226,663 held-out slices; single fit, no pooling). **(A)** Slice-level area under the receiver operating characteristic curve (AUC) joined to patient-level AUC after mean aggregation for each of the six official labels, with 95% patient-clustered bootstrap intervals (2000 replicates) and the within-series label permutation null at the patient unit. Nothing changes between the two readings of a pair except the unit at which the ranking is performed. Three patient-level intervals lie wholly below chance and the other three cross it; epidural, on 108 positive patients, is the only point estimate above chance. The patient-level null lies above chance because permuting within a series preserves stack depth. **(B)** The mechanism, for the "any hemorrhage" label: the same patient-level score read unstratified, within exact stack depth, and within 5-slice depth strata, with stack depth read alone for comparison. Holding depth fixed removes the sub-chance reading; the depth-conditioned value of 0.5071 (95% CI: 0.5013, 0.5129) excludes 0.500 but sits 0.0016 above the within-series permutation null computed at fixed depth, 0.5055 (range, 0.5035 to 0.5068), shown as the shaded band, which its own interval spans. **(C)** The same four quantities over the 24 independently drawn holdouts of the identical design that furnish the primary estimate. Points are the individual holdouts, the bar is the family range, and the open marker is the pre-registered holdout of panels A and B, which sits at the edge of the family on the patient-level and depth-conditioned readings; that is why the family mean and not this draw is reported as primary. The constant predictor is exactly 0.500 in every holdout, which is the property the superseded pooled out-of-fold estimator did not have.

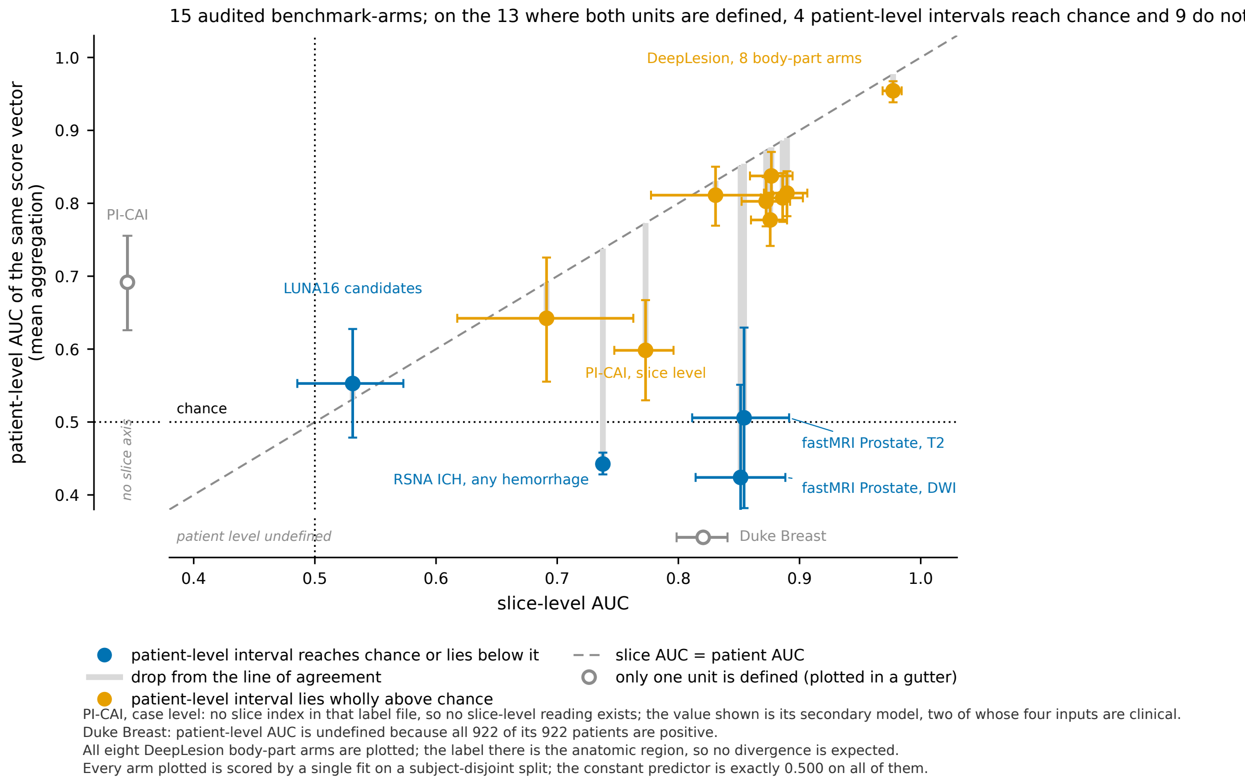

Figure 2. The same pixel-blind score vector read at two units, for every audited benchmark-arm. Each point is one arm: its slice-level area under the receiver operating characteristic curve (AUC) against the patient-level AUC of the identical score vector after mean aggregation, with 95% subject-clustered bootstrap intervals on both axes (replicate counts in Table 4). The dashed diagonal is the line on which the two units agree, and the grey stem below each point is the distance that arm falls from it. All eight DeepLesion body-part arms are plotted, not one of them: they are the arms on which no divergence occurs, and that is expected rather than anomalous, because the label there is the anatomic region and relative slice position predicts it at either unit. Values are for the locked 20-bin positional baseline except PI-CAI, whose positional baseline is exactly 0.500 on the public marksheet, which carries no slice index, and whose secondary tree over that file's four non-image columns—two of them clinical, age and prostate-specific antigen level—is drawn instead. That benchmark's slice-level arm, read from its 1,295 lesion delineation volumes under the same locked protocol, is plotted separately and at both units. Every arm plotted is scored by a single fit on a subject-disjoint split and its constant predictor is exactly 0.500. The two arms defined at only one unit are drawn in the margins rather than in the field, so that a value that does not exist cannot be read as a low one: PI-CAI's label file carries no slice index, and Duke Breast's patient-level AUC is undefined because all 922 of its 922 patients are positive.

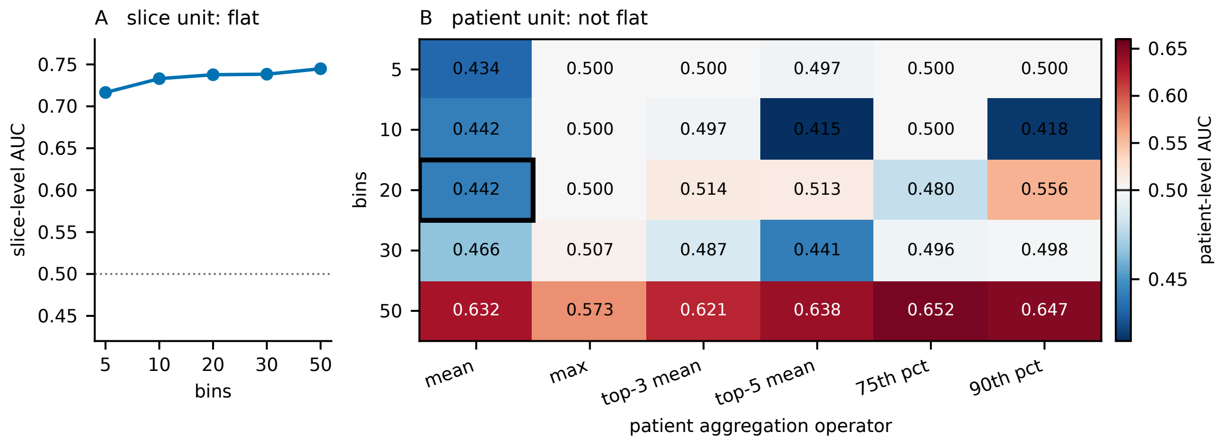

Figure 3. How the two units respond to the two analysis choices the positional baseline requires. **(A)** Slice-level AUC against the number of position bins: 0.717, 0.733, 0.738, 0.738 and 0.745 over 5, 10, 20, 30 and 50 bins, a range of 0.028. **(B)** Patient-level AUC over the same bin counts crossed with six aggregation operators, on the same frozen holdout of 5,681 patients and the same single fit. The boxed cell, 20 bins with mean aggregation, is the pre-specified analysis reported throughout; every other cell is a sensitivity analysis and none was used to choose it. Colour is centred on 0.500. The patient reading spans 0.415 to 0.652, a range of 0.237, against 0.028 at the slice unit for the identical fits. Two features carry the paper's caveat rather than its claim: at 50 bins every operator reads above chance, because a bin then holds about one slice and the aggregate begins to track stack depth directly; and the operators that read near exactly 0.500 are degenerate, taking only a handful of distinct values across all 5,681 patients. The slice-level reading is nearly invariant to both choices. The patient-level reading is not, and any patient-level statement made here is a statement about mean aggregation at 20 bins.

Supplemental Material

What a Slice-Level Benchmark Certifies without the Pixels: An Audit of Six Public Imaging Datasets

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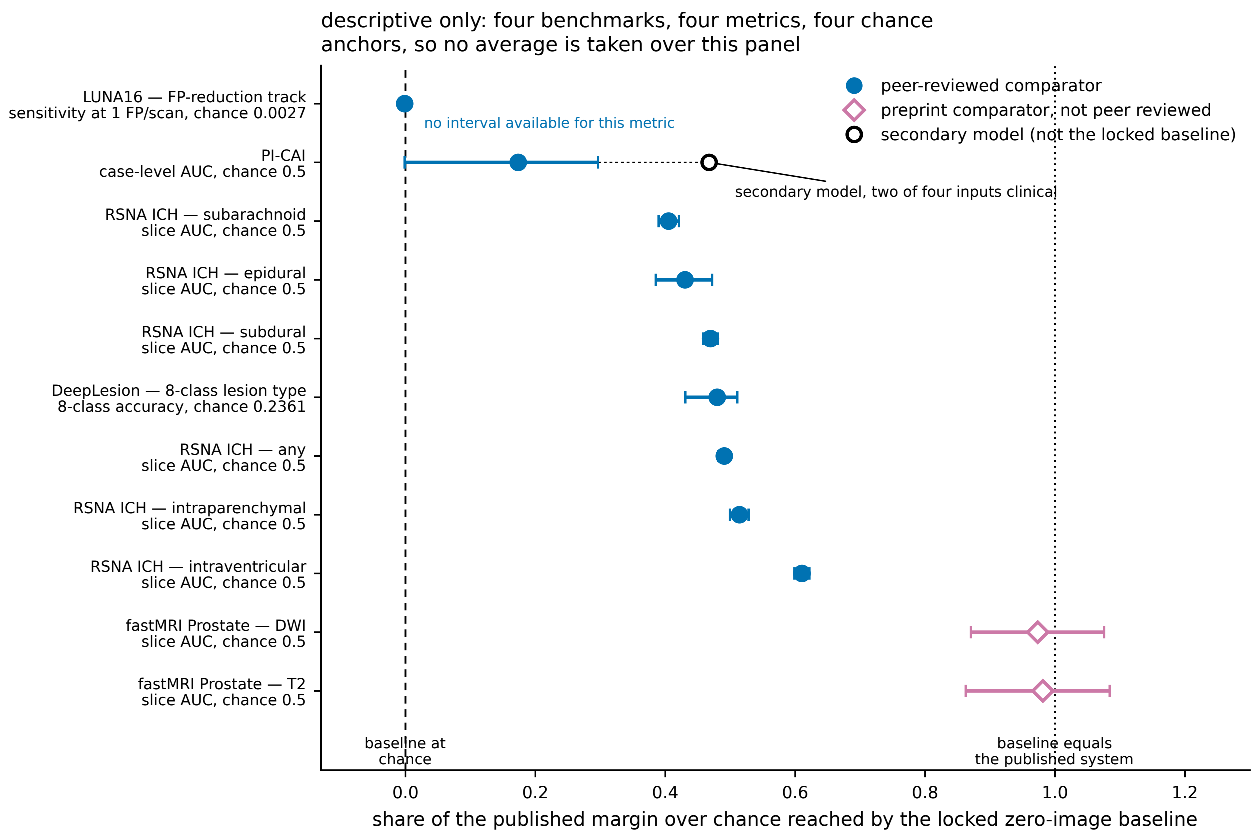

Figure S1. A descriptive cross-study comparison: the share of a published margin over chance reached by the locked pixel-blind baseline, for the strongest published comparator on each benchmark. This is not an inferential quantity and no average is taken over the panel. The four peer-reviewed benchmarks sit on four different metrics against four different chance anchors, printed on each row, and they run from about half the published margin to none of it; that spread is the finding. Error bars are 95% subject-clustered bootstrap intervals propagating uncertainty in the baseline only. Rows whose comparator is a preprint that has not been peer reviewed are marked as such and are the two highest here, at 0.981 and 0.973. LUNA16 is plotted at -0.001, computed on the frozen holdout from sensitivity at one false positive per scan and not from the challenge's competition performance metric; the reference used in place of 0.5 is the random-score value of that metric on the same held-out scans. No interval is available for that row. PI-CAI is plotted at 0.174, read at the case unit from the locked positional baseline fitted on its lesion delineation volumes and mean aggregated. An earlier revision reported this row as not estimable, which was true of that benchmark's public marksheet—it carries no slice index—and false of the benchmark, whose delineation volumes supply one. That row is across cohorts: the published comparator is on the hidden test set and the baseline is on the development set. The open marker on the same row is its secondary tree over four non-image columns, two of which are clinical predictors rather than acquisition fields, and is not part of this comparison.